

OOP Project Task: Hospital Management System (CSV Based)

Objective

Create a Hospital Management System using **OOP (Object-Oriented Programming)** where all patient data is stored in a **CSV file**.

Classes

1) Patient Class

Attributes

- patient_id
- name
- age
- gender
- disease
- doctor_name
- admission_date
- bill_amount

Methods

1. add_patient()
 - Add new patient record to CSV.
2. view_all_patients()
 - Display all patients.
3. search_patient(patient_id)
 - Search patient using Patient ID.
4. update_patient(patient_id)
 - Update patient details.
5. delete_patient(patient_id)
 - Remove patient record.
6. calculate_bill(patient_id)
 - Display bill amount.
7. discharge_patient(patient_id)
 - Mark patient as discharged.

8. count_patients()
 - Show total patients.
 9. show_patients_by_doctor(doctor_name)
 - Display all patients under a specific doctor.
 10. show_patients_by_disease(disease)
 - Display all patients having a specific disease.
 11. highest_bill_patient()
 - Show patient with highest bill.
 12. average_bill()
 - Calculate average bill of all patients.
 13. export_summary()
 - Generate summary report.
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2) Hospital Class

Methods

1. menu()
 - Display menu options.
 2. hospital_info()
 - Show hospital details.
 3. total_revenue()
 - Calculate total revenue from all bills.
 4. doctor_wise_patient_count()
 - Count patients under each doctor.
 5. generate_daily_report()
 - Generate daily report.
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CSV File Structure

patient_id,name,age,gender,disease,doctor_name,admission_date,bill_amount,status

101,Vaibhav,25,Male,Fever,Dr.Sharma,2025-06-10,5000,Admitted
102,Rahul,30,Male,Malaria,Dr.Patil,2025-06-11,8000,Discharged